



A.M.($\leq$)	Any	Moon P.	Gray
Image Quality		Seeing < 1.5"	
Cloudy Cover		Photometric	

Title of Proposal: Rotation Photometry of the Trojan object (1143) Odysseus

Telescope 0.60m Boller & Chivens

Abstract:

The Trojan asteroids of Jupiter carry valuable information about the Solar System and its formation. Their physical characteristics are closely related to the dynamical evolution of this gas giant. (1143) Odysseus is a Trojan located at the Lagrange point L4, with an equivalent diameter of approximately 120 km. Stellar occultations by this object were observed and analyzed in 2023 and 2024, revealing that the Trojan has an irregular shape. For the year 2026, a new occultation is predicted for May 7th, for which we will organize an observational campaign. In this project, we intend to obtain the rotational light curve of this object on a date close to the predicted occultation, in order to determine the rotational phase of the body at the time of the event and thus better understand its true three-dimensional shape through the combined use of both techniques.

Time requested: 44 hours.

Minimum time accepted: 32 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

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Is there a thesis or dissertation that will benefit from the data?: YES

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Optimal dates:

From April 10 to 13, May 8 to 10.

Impossible dates:

Before April 10, on May 7, and after May 19.

Instruments:

Instrument	Camera
Imager (CAM 1; 2 or 4)	Ikon 2048 x 2048

Instrument	Filter	Grating	Slit
Imager (CAM 1; 2 or 4)			
	UBVRI		

Scientific Justification

Trojan objects play a crucial role in our understanding of the dynamical evolution of the gas giants. Odysseus has been observed and studied through the stellar occultation technique since 2023. One of its latest events, in February of 2024, provided results that indicate an irregularity. In this event, 6 positive chords and 2 negative chords were detected.

The analysis from this occultation allows us to infer a non-regular shape for (1143) Odysseus, as the elliptical adjustment was unable to describe the shape of the Trojan. This urges a more profound analysis of the object, mainly by obtaining its rotation light curve close to the date of a future occultation event. In the literature, we find some rotation curves for the object (Stephens, 2023), where the rotation period is estimated at 10.124 ± 0.006 hours, as shown in Figure 1.

Observations of rotation light curves near stellar occultation events enable us to determine the rotation phase of the object and the area visible to the observer during the event. In the case of Odysseus, new observations will provide a more precise value for the rotation period, helping to determine its rotation phase for a close occultation event.

The prediction map of a future occultation by Odysseus can be seen in Figure 2, which will involve a bright star, making the observation easier for amateur astronomers in the region, who already cooperate with our work, favoring a good coverage, and thus, new measurements of the object's shape

Therefore, we intend to obtain precise rotational light curves of the object during four nights of observations, starting from April 10, when it will be observable for over 8 hours in a single night. With that, we will be able to cover the full rotation period (≈ 10.12 hours) during this run.

We also request two nights in May to better determine the rotation phase after the occultation event on May 7. The ideal would be the nights after May 7. We will utilize the light curve inversion technique, in conjunction with data already available in the literature, to refine a 3D model of the celestial body. This model's surface is represented by a set of polyhedra that generate a good approximation of the real shape of the object (Kaasalainen et al. 2001).

References

1. Kaasalainen, Torppa & Muinonen 2001 *Icarus*, 153, 37.
2. Nesvorný, Vokrouhlický & Morbidelli 2013 *ApJ*, 768, 45.
3. Mottola et al. 2024 *Space Sci. Rev.*, 220, 17.
4. Grav et al. 2011 *ApJ*, 742, 40.
5. Mottola et al. 2011 *AJ*, 141, 170.
6. Kaasalainen & Torppa 2001 *Icarus*, 153, 24.
7. Elliot 1979 *ARA&A*.
8. Viikinkoski, Kaasalainen & Durech 2015 *A&A*, 576, A8.
9. Morbidelli et al. 2005 *Nature*, 435.
10. Stephens, Robert D. 2023 *Minor Planet Bulletin*, 157, 50.

Experimental Design

The target of this proposal is the Trojan (1143) Odysseus, a study about Odysseus's shape and size has been in progress for the past two years following the PI's masters degree, which will be completed in 2026. The data from this observation will be fundamental to the paper related to the dissertation associated with this work.

The purpose of the observation is to obtain rotation light curves of the target body on different nights in an attempt to capture most of its rotation period of 10.124 ± 0.006 hours.

The observable of this observation will be the light curve of the object, so that the variation of the flux in the light curve may not only help determine the rotation phase of Odysseus close to the next stellar occultation event, but also allow for the use of the light curve inversion technique to better refine the first estimate for a 3D model of the Trojan.

To achieve the expected results, the light curve from this observation will be added to the dataset used for studying the object, where both rotation light curves and occultation chords will be fed into the ADAM software, which is used to generate a 3D model of the body. With the lightcurve inversion technique, this model will be used to generate synthetic lightcurves with the goal of matching the observed data, for this step minimization techniques such as chi-squared and RMSD are combined with genetic algorithm to find the best adjustment for the model.

Technical Description

Since the goal is to obtain the rotational light curve of Odysseus, which has an apparent magnitude of approximately 16, observations should not be carried out during full Moon nights, in order to improve the signal-to-noise ratio. New Moon or quarter phases are suitable for this case.

Given that the rotation period of Odysseus is 10.12 hours, four nights (with a minimum of three nights), preferably starting from April 10, are required to cover its full rotation period. The object will be observable for just over 8 hours per night (at altitudes above 30 degrees), justifying the need for the requested hours to cover a full rotation period (≈ 10.12 hours). Additionally, two nights after May 8 will allow us to determine the rotation phase near the occultation event on May 7. Important to note that during this epoch (April), the object will be observable for only 6 hours or less during one night; thus, observations during this period are not ideal to obtain a full light curve coverage.

Based on our past observations, for an object of 16 magnitude, an exposure time of approximately 90 seconds will provide SNR over 50. This should provide a photometric precision of about 0.02 magnitudes, which is enough to obtain a good light curve, considering that the expected amplitude exceeds 0.4 magnitudes.

Previous missions of this proposal to this telescope

There are no previous missions for this proposal

Previous missions of other proposals to this telescope

- Studying distant small Solar System bodies through stellar occultations - bright stars. All data reduced. Positions were obtained and sent to <https://lesia.obspm.fr/lucky-star/predictions.php>. The results from proposals like this are important to improve the predictions of stellar occultations.
- Rotation Photometry of the Trojan object (3451) Mentor. All data reduced. Positions were obtained and sent to <https://lesia.obspm.fr/lucky-star/predictions.php>. The results from proposals like this are important to improve the predictions of stellar occultations.

Previous results in the field by the Principal Investigator

1. RODRIGUES, J. A. N.; RIBAS, F. B. ; FREDERIC, V. ; DESMARS, J. ; MARGOTI, G. ; PEREIRA, C. L. ; MORGADO, B. E. ; ORTIZ, J. L. ; SICARDY, B. ; FERNANDEZ-VALENZUELA, E. ; HOLLER, B. ; KRETLOW, M. . Ocultações Estelares pelo satélite de Quaoar (50000/1) Weywot, XII Taller de Ciências Planetárias, 2024.
2. RODRIGUES, J. A. N.; RIBAS, F. B. ; DESMARS, J. ; MARGOTI, G. ; ORTIZ, J. L. ; SICARDY, B. ; KRETLOW, M. . Stellar Occultations by the Trojan 1143 Odysseus, XLVII Reunião da Sociedade Astronômica Brasileira, 2024
3. RODRIGUES, J. A. N.; RIBAS, F. B. ; DESMARS, J. ; MARGOTI, G. ; GRADOVSKI, E. ; ORTIZ, J. L. ; SICARDY, B. ; KRETLOW, M. . Determining the size and shape of the Trojan (1143) Odysseus through stellar occultation technique, reverse light curve analysis and 3D modeling, XXII Brazilian Colloquium on Orbital Dynamics, 2024.

PI and Co-Is publications relevant to this proposal

1. The Rings of (2060) Chiron: Evidence of an Evolving System, Pereira, C.L. et al. (2025), ApJL, DOI: 10.3847/2041-8213/ae0b6d
2. Size and Shape of Jupiter Trojan (2207) Antenor from Stellar Occultations, Ferreira, F.S. et al. (2025), AJ, DOI: 10.3847/1538-3881/ade0dc
3. The trans-Neptunian object (119951) 2002 KX14 revealed via multiple stellar occultations, Rizos, J.L et al. (2025), A&A, DOI: 10.1051/0004-6361/202554154
4. Centaur 29P/Schwassmann-Wachmann 1 and its near-nucleus environment from a stellar occultation, Pereira, C.L. et al. (2025), Philosophical Transactions A, DOI: 10.1098/rsta.2024.0189
5. A study of centaur (54598) Bienor from multiple stellar occultations and rotational light curves, Kretlow, M. et al. (2024), A&A, DOI: 10.1051/0004-6361/202450833
6. Constraints on the evolution of the Triton atmosphere from occultations: 1989-2022, Sicardy, B. et al. (2024), A&A, DOI: 10.1051/0004-6361/202348756
7. Physical properties of Centaur (60558) 174P/Echeclus from stellar occultations, Pereira, C.L. et al. (2024), MNRAS, DOI: 10.1093/mnras/stad3318
8. Kilometer-precise (Ull) Umbriel physical properties from the multichord stellar occultation on 2020 September 21, Assafin, M. et al. (2023), MNRAS, DOI: 10.1093/mnras/stad3093
9. A large topographic feature on the surface of the trans-Neptunian object (307261) 2002 MS4 measured from stellar occultations, Rommel, F. et al. (2023), A&A, DOI: 10.1051/0004-6361/202346892

10. *Changing material around (2060) Chiron revealed by an occultation on December 15, 2022*, Ortiz, J.L. et al. (2023), A&A, DOI: 10.1051/0004-6361/202347025
11. Constraints on (2060) Chiron's size, shape, and surrounding material from the November 2018 and September 2019 stellar occultations, Braga-Ribas, F. et al. (2023), A&A, DOI: 10.1051/0004-6361/202346749
12. The two rings of (50000) Quaoar, Pereira, C.L. et al. (2023), A&A, DOI: 10.1051/0004-6361/202346365
13. A dense ring of the trans-Neptunian object Quaoar outside its Roche limit, Morgado, B.E. et al. (2023), Nature, DOI: 10.1038/s41586-022-05629-6
14. The multichord stellar occultation by the centaur Bienor on January 11, 2019, Fernández-Valenzuela, E. et al. (2023), A&A, DOI: 10.1051/0004-6361/202243214
15. Physical properties of the trans-Neptunian object (38628) Huya from a multi-chord stellar occultation, Santos-Sanz, P. et al. (2022), A&A, DOI: 10.1051/0004-6361/202141546
16. The multichord stellar occultation on 2019 October 22 by the trans-Neptunian object (84922) 2003 VS₂, Vara-Lubiano, M. et al. (2022), A&A, DOI: 10.1051/0004-6361/202141842
17. Constraints on the structure and seasonal variations of Triton's atmosphere from the 5 October 2017 stellar occultation and previous observations, Marques Oliveira, J. et al. (2022), A&A, DOI: 10.1051/0004-6361/202141443
18. Refined physical parameters for Chariklo's body and rings from stellar occultations observed between 2013 and 2020, Morgado, B.E. et al. (2021), A&A, DOI: 10.1051/0004-6361/202141543
19. The 2017 May 20 stellar occultation by the elongated centaur (95626) 2002 GZ32, Santos-Sanz, P. et al. (2021), MNRAS, DOI: 10.1093/mnras/staa3881
20. Stellar occultations enable milliarcsecond astrometry for Trans-Neptunian objects and Centaurs, Rommel, F.L. et al. (2020), A&A, DOI: 10.1051/0004-6361/202039054
21. A multi-chord stellar occultation by the large trans-Neptunian object (174567) Varda, Souami, D. et al. (2020), A&A, DOI: 10.1051/0004-6361/202038526
22. The large trans-Neptunian object 2002 TC302 from combined stellar occultation, photometry, and astrometry data, Ortiz, J.L. et al. (2020), A&A, DOI: 10.1051/0004-6361/202038046
23. The first observed stellar occultations by the irregular satellite Phoebe (Saturn IX) and improved rotational period, Gomes-Júnior et al. (2020), MNRAS, DOI: 10.1093/mnras/stz3463
24. Stellar occultation by TNOs: from predictions to observations, Ortiz, J.L. et al. (2020), *The Trans-Neptunian Solar System*, Edited by Dina Prialnik, Maria Antoinetta Barucci, Leslie Young. ISBN: 9780128164907. Elsevier, p.413-437, DOI: 10.1016/B978-0-12-816490-7.00019-9
25. The trans-Neptunian object (84922) 2003 VS₂ through stellar occultations, Benedetti-Rossi et al. (2019), AJ, DOI: 10.3847/1938-3881/ab3b05
26. First stellar occultation by the Galilean moon Europa and upcoming events between 2019 and 2021, Morgado, B. et al. (2019), A&A, DOI: 10.1051/0004-6361/201935500
27. Pluto's ephemeris from ground-based stellar occultations (1988-2016), Desmars, J., Meza, E., Sicardy, B., Assafin, M., Camargo, J.I.B. et al. (2019), A&A, DOI: 10.1051/0004-6361/201834958
28. Lower atmosphere and pressure evolution on Pluto from ground-based stellar occultations, 1988-2016, Meza, E. et al. (2019), A&A, DOI: 10.1051/0004-6361/201834281
29. Astrometry and Occultation Predictions to Trans-Neptunian and Centaur Objects Observed within the Dark Energy Survey, Banda-Huarca, M. V., Camargo, J.I.B., Desmars, J. et al. (2019), AJ, DOI:10.3847/1538-3881/aafb37
30. The future of stellar occultations by distant solar system bodies: Perspectives from the Gaia astrometry and the deep sky surveys, Camargo, J. I. B. et al. (2018), P&SS, DOI: 10.1016/j.pss.2018.02.014
31. The size, shape, density and ring of the dwarf planet Haumea from a stellar occultation, Ortiz, J.L., et al. (2017), Nature, DOI:10.1038/nature24051

32. The Structure of Chariklo's Rings from Stellar Occultations, Bérard, D., Sicardy, B., Camargo, J. I. B., et al. (2017), AJ, DOI:10.3847/1538-3881/aa830d
33. Size and shape of Chariklo from multi-epoch stellar occultation, Leiva, R., Sicardy, B., Camargo, J. I. B., et al. (2017), AJ, DOI:10.3847/1538-3881/aa8956
34. Study of the plutino object (208996) 2003 AZ84 from stellar occultations: size, shape and topographic features, Dias-Oliveira, A., Sicardy, B., Ortiz, J. L., Leiva, R., Braga-Ribas, F., Vieira-Martins, R., Benedetti-Rossi, G., Camargo, J. I. B. et al. (2017), AJ, DOI: 10.3847/1538-3881/aa74e9
35. Results from a triple chord stellar occultation and far-infrared photometry of the trans-Neptunian object (229762) 2007 UK126, Schindler, K. et al. (2017), A&A, DOI: 10.1051/0004-6361/201628620
36. Results from the 2014 November 15th Multi-chord Stellar Occultation by the TNO (229762) 2007 UK126, Benedetti-Rossi, B. et al. (2016), AJ, DOI:10.3847/0004-6256/152/6/156
37. Pluto's Atmosphere from the 2015 June 29 Ground-based Stellar Occultation at the Time of the New Horizons Flyby, Sicardy, B. et al. (2016), ApJ, DOI:10.3847/2041-8205/819/2/L38
38. Pluto's Atmosphere from Stellar Occultations in 2012 and 2013, Dias-Oliveira, A. et al. (2015), ApJ, DOI:10.1088/0004-637X/811/1/53
39. Results of two multichord stellar occultations by dwarf planet (1) Ceres, Gomes-Junior, A.R. et al. (2015), MNRAS, DOI:10.1093/mnras/stv1081
40. A ring system detected around the Centaur (10199) Chariklo, Braga-Ribas et al. (2014), Nature, DOI: 10.1038/nature13155
41. An exploration of Pluto's environment through stellar occultations, Boissel et al. (2014), A&A. DOI: 10.1051/0004-6361/201321836
42. Candidate stellar occultations by Centaurs and trans-Neptunian objects up to 2014, Camargo et al. (2014), A&A, DOI:10.1051/0004-6361/201322579
43. The Size, Shape, Albedo, Density, and Atmospheric Limit of Transneptunian Object (50000) Quaoar from Multi-chord Stellar Occultations, Braga-Ribas et al. (2013), ApJ, DOI: 10.1088/0004-637X/773/1/26
44. Albedo and atmospheric constraints of dwarf planet Makemake from a stellar occultation, Ortiz, J.L. et al. (2012), Nature, DOI:10.1038/nature11597
45. Obtenção dos parâmetros rotacionais e escala de troianos de Júpiter através de curvas de rotação e ocultações Gradovski, E., Dissertação (Mestrado em Física e Astronomia) - Universidade Tecnológica Federal do Paraná, Curitiba, 2025.

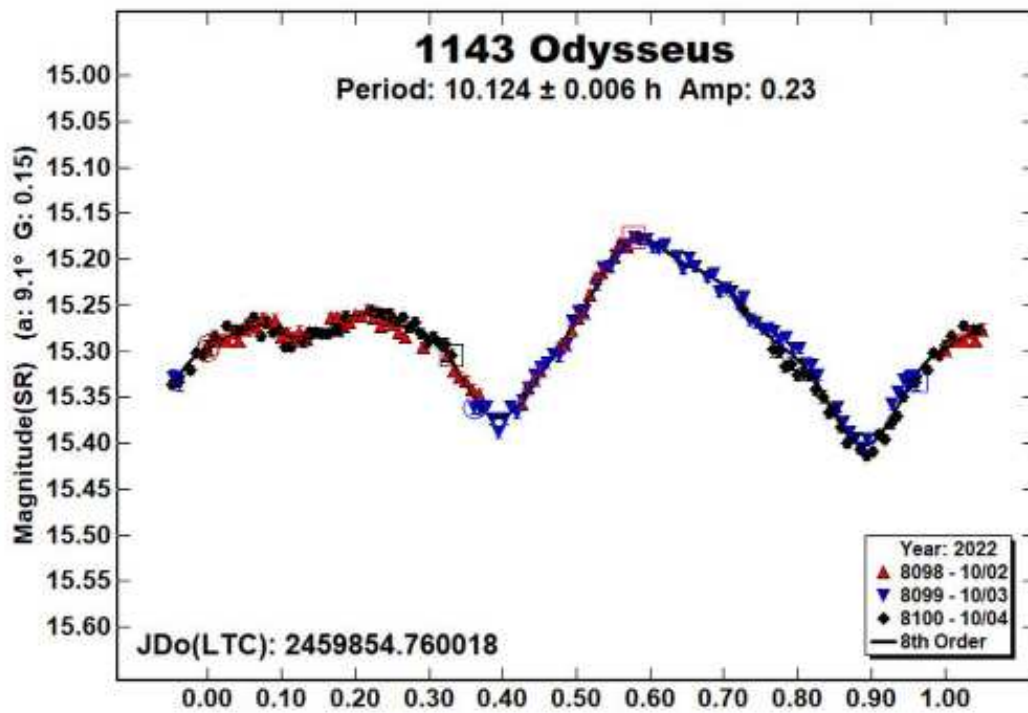


Figure 1: 1143 Odysseus lightcurve obtained by Robert D. Stephens in 2022 (Stephens, 2023)

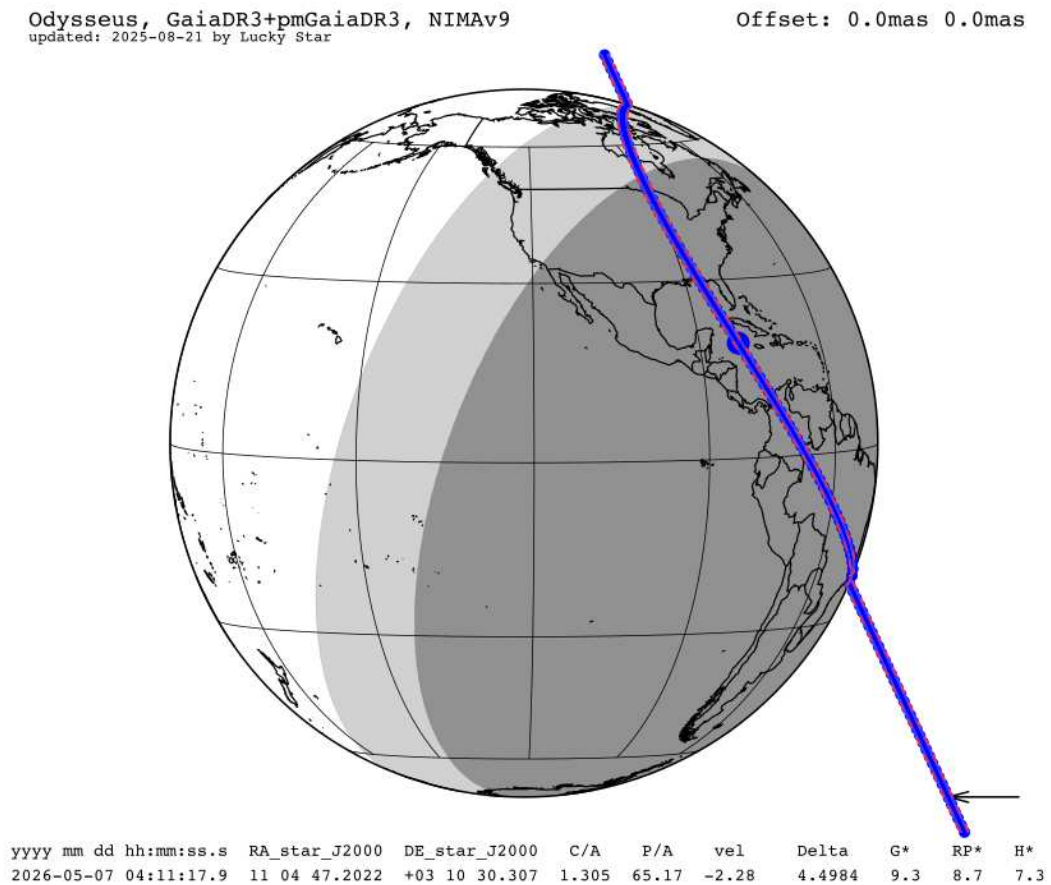


Figure 2: Prediction map for the stellar occultation by (1143) Odysseus to take place on May 7th, 2026.

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
(1143) Odysseus	11:09:34.21	+02:22:09.4	15.504	G	Sci	Values for astrometric position and magnitude for 10/APR
(1143) Odysseus	11:04:45.51	+03:11:58.2	15.846	G	Sci	Values for astrometric position and magnitude for 08/MAY